

Voltage Amplifier Design Project

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1. Design Summary

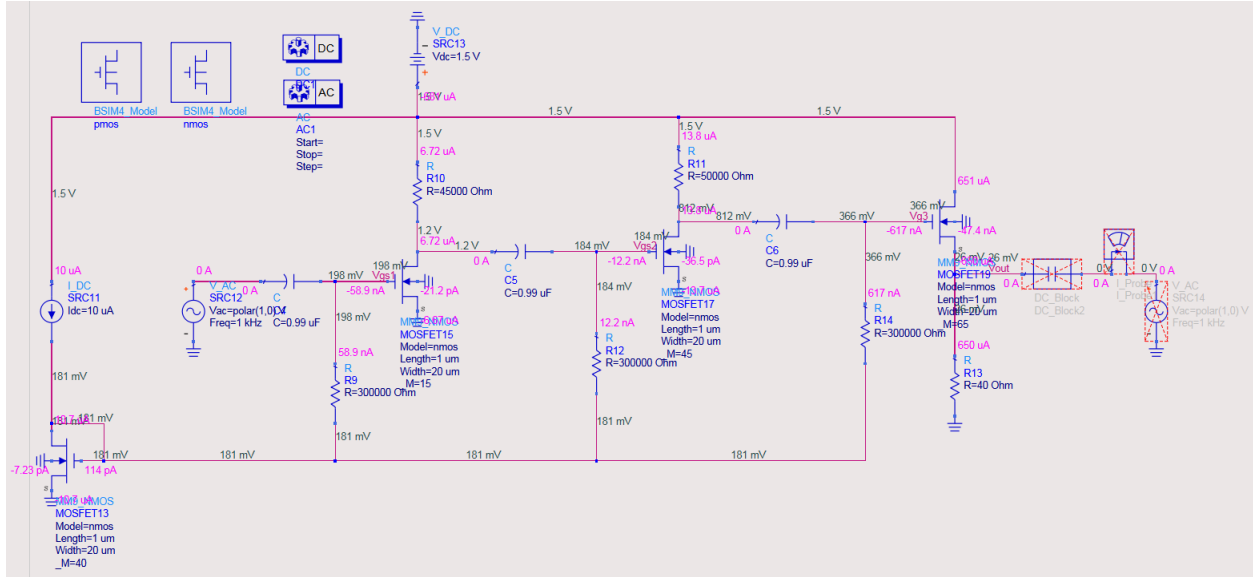
My project utilizes a three-stage amplifier configuration consisting of two CMOS Common Source stages followed by a CMOS Common Drain stage. This configuration is chosen over a traditional two-stage Common Source-Common Drain amplifier since it offers higher voltage gain while maintaining benefits such as optimal input and output impedance. The Common Source stage provides high input impedance (close to infinity) while the Common Drain stage offers low output impedance (close to zero), enhancing overall performance of the voltage amplifier.

2. Performance Comparison Table

	Target Specifications	Simulated Performance
VDD	$\leq 1.5 \text{ V}$	1.5 V
Frequency	1 kHz	1 kHz
Total DC Current*	$\leq 2 \text{ mA}$	650 μA
Magnitude of Voltage Gain	$\geq 20 \text{ dB}$	27.5672 dB
Input Impedance	$\geq 100 \text{ k}\Omega$	295598.1994 Ω
Output Impedance	$\leq 30 \text{ }\Omega$	24.1055 Ω
Reference Current of Current Mirrors	10 μA	10 μA

*Total DC Current was seen directly from the V_{DD} supply voltage using pin current annotation on the Keysight ADS software.

3. Complete Schematic



4. Theoretical Analysis

I. Theoretical Voltage Gain

$$A_v = g_{m1} R_{D1} \times g_{m2} R_{D2} \times \frac{R_s}{\left(\left(\frac{1}{g_{m3}}\right) + R_s\right)}$$

$$= ((1.322 \times 10^{-4}) \times (45000)) \times ((2.485 \times 10^{-4}) \times (50000)) \times \frac{40}{\left(\left(\frac{1}{0.012}\right) + 40\right)} = 23.9729$$

$$dB = 20 \log_{10}(A_v) = 20 \log_{10}(23.9729) = 27.5944$$

II. Theoretical Input Impedance

$$R_{in} = \frac{\infty \times R_G}{\infty + R_G} = \frac{\infty \times 300000}{\infty + 300000} = 300000$$

III. Theoretical Output Impedance

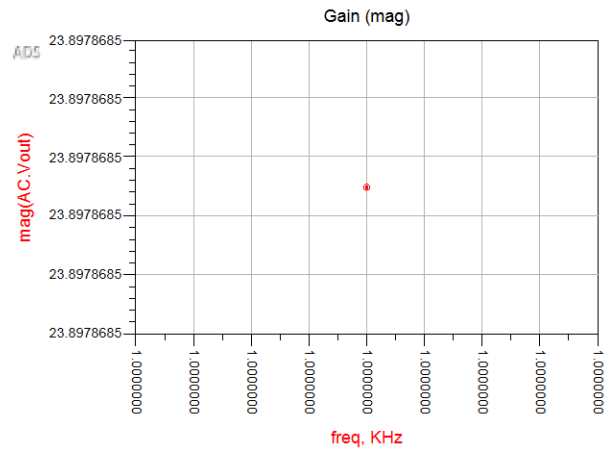
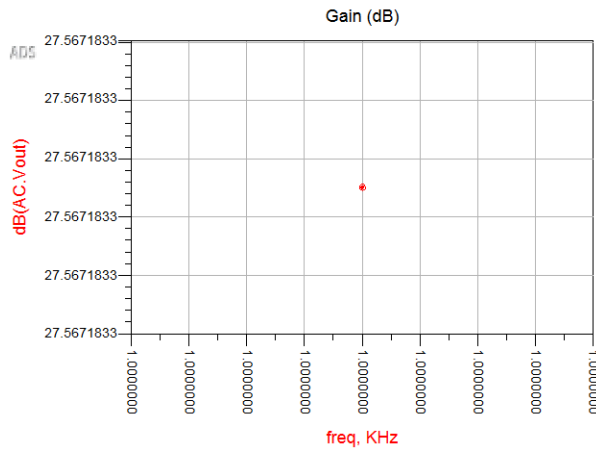
$$R_{out} = \frac{R_s}{g_{m3} R_s + 1} = \frac{40}{0.012(40) + 1} = 27.0270$$

5. Simulation Results

I. Simulated Voltage Gain

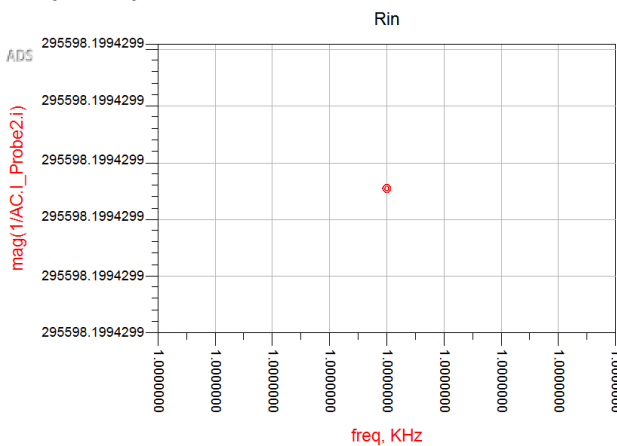
A_v (magnitude) = 23.8979

A_v (dB) = 27.5672

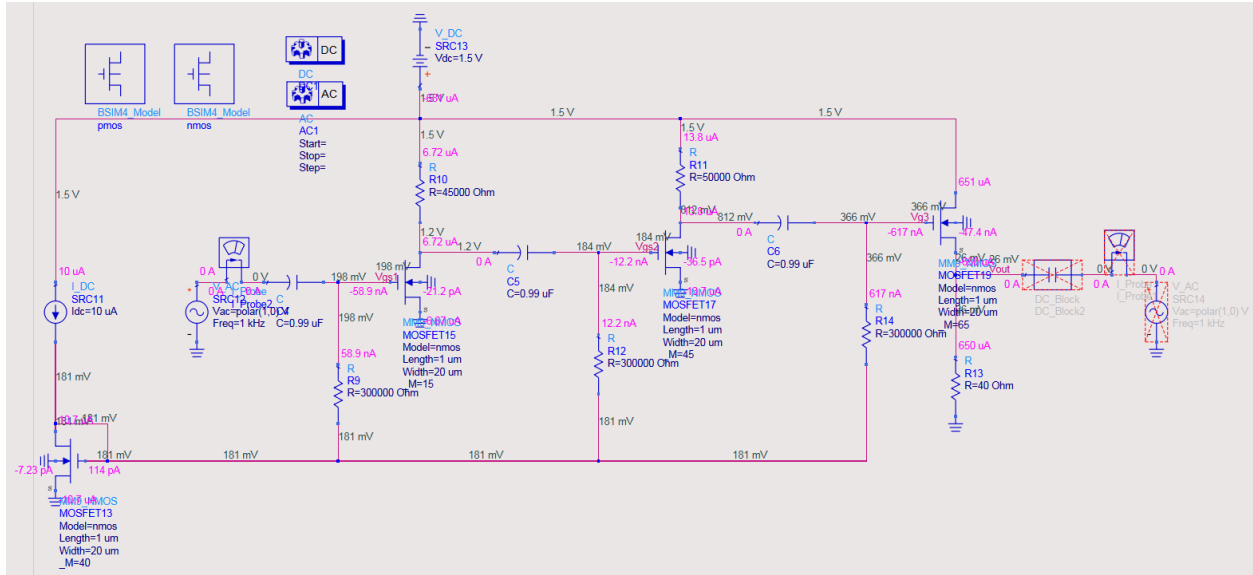


II. Simulated Input Impedance

R_{in} (ohms) = 295598.1994

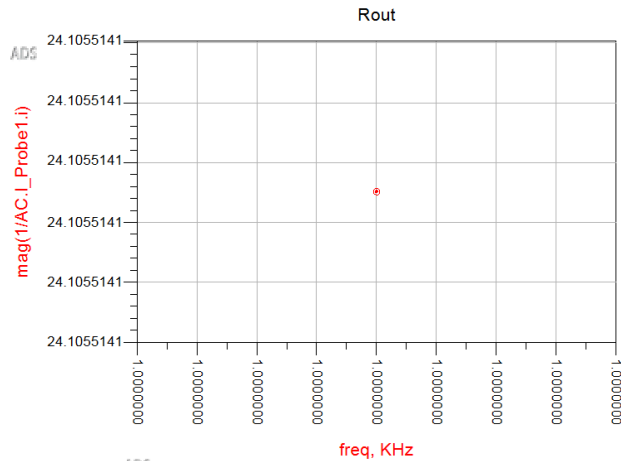


Procedure: To measure the input impedance, a current probe was placed right after the AC voltage source at the input of the amplifier. Then the ADS simulation plot function was used to calculate the magnitude of 1 over the measured current at the probe. This circuit setup is shown below.

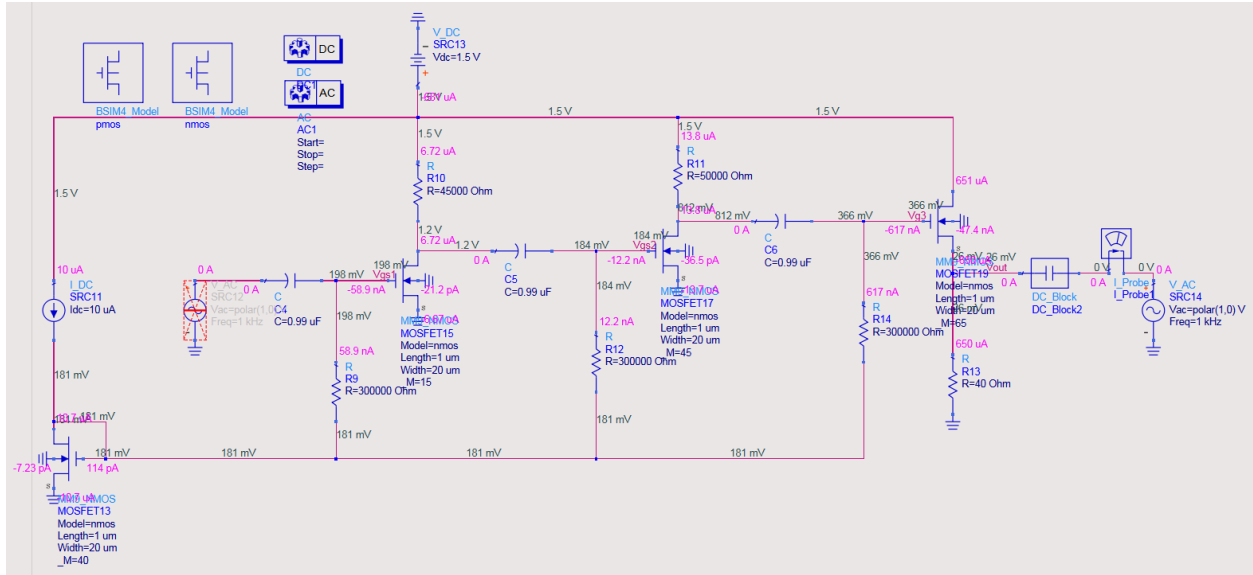


III. Simulated Output Impedance

$$R_{out} \text{ (ohms)} = 24.1055$$



Procedure: To measure the output impedance, the AC voltage source at the input was first deactivated and shorted. A DC block, current probe, and AC voltage probe were placed in the same order at the output. Then the ADS simulation plot function was used to calculate the magnitude of 1 over the measured current at the probe. This circuit setup is shown below.



IV. Simulated Transconductance

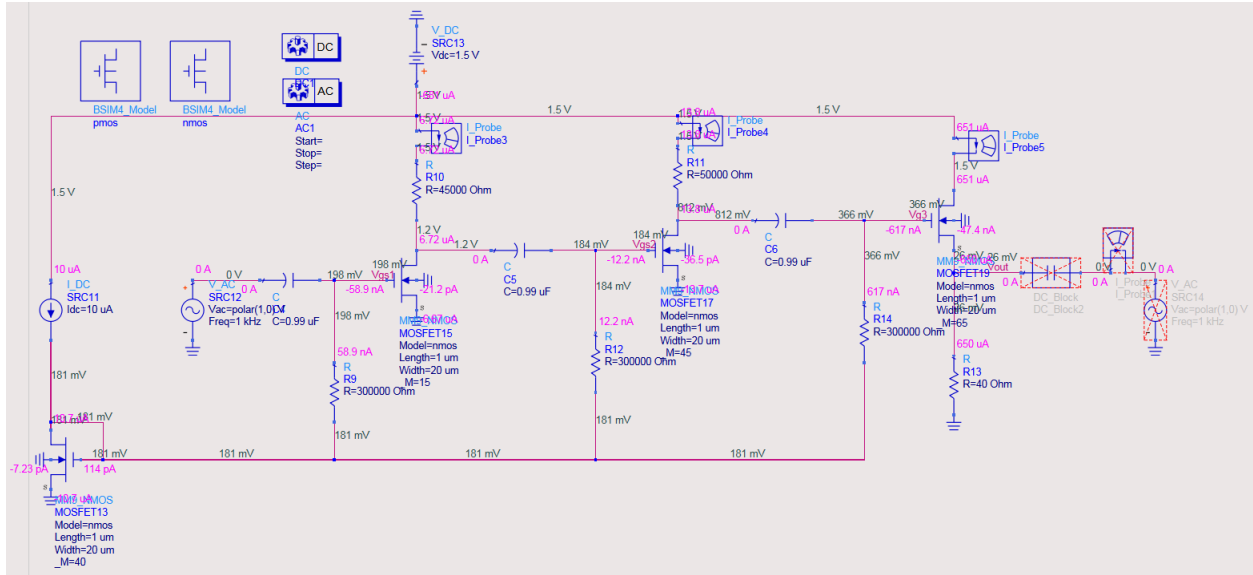
$$g_{m1} = 1.322 \times 10^{-4}$$

$$g_{m2} = 2.485 \times 10^{-4}$$

$$g_{m3} = 0.012$$

freq	AC.I_Probe3.i/AC.Vgs1	AC.I_Probe4.i/AC.Vgs2	AC.I_Probe5.i/(AC.Vg3-AC.Vout)
1.000 kHz	1.322E-4 / -0.112	2.485E-4 / -0.097	0.012 / -3.731E-5

Procedure: To measure the three transconductances, a current probe was placed at each transistor's drain to measure the I_{ds} current. The V_{gs} voltage for the Common Source stages, or the V_g voltage for the Common Drain stage, was labeled at each transistor's gate terminal. Then the ADS simulation list function was used to calculate I_{Probe} over V_{gs} for the CS stages, and I_{Probe} over $(V_g - V_{out})$ for the CD stage. This circuit setup is shown below.



V. Comparison between Calculated and Simulated Performance

	Calculated	Simulated
Voltage Gain (magnitude)	23.9729	23.8979
Voltage Gain (dB)	27.5944	27.5672
Input Impedance (ohms)	300000	295598.1994
Output Impedance (ohms)	27.0270	24.1055

I. Voltage Gain Accuracy

$$\text{Percent Error (magnitude): } \frac{|23.8979 - 23.9729|}{23.9729} \times 100 = 0.3128\%$$

$$\text{Percent Error (dB): } \frac{|27.5672 - 27.5944|}{27.5944} \times 100 = 0.0986\%$$

II. Input Impedance Accuracy

$$\text{Percent Error: } \frac{|295598.1994 - 300000|}{300000} \times 100 = 1.4673\%$$

III. Output Impedance Accuracy

$$\text{Percent Error: } \frac{|24.1055 - 27.0270|}{27.0270} \times 100 = 10.8096\%$$

IV. Performance Difference Explanation

The above percent error calculations indicate that the voltage amplifier circuit is quite accurate since all the parameters had a difference under about 10% between the simulated and calculated values. Some of the performance disparities may be due to random simulation errors and parasitic capacitances, as well as electronic parts not being perfectly ideal devices in practice. Also, the sizing of the transistors, including length, width, and number of fingers, could affect the voltage and current measurements at their terminals, such as in the output impedance measurement.